

Operator reformulation of the off-hook order law rank-flow, the first-pinch collapse, and the merged-junction gap

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Abstract

The order law $\text{ord}_{x=q^2} Z_\lambda = \binom{\tau+1}{2}$ for the Baxterised staircase monodromy on the Hoefsmit module V^λ was treated at the trace level: the within-arc reach lemma was proved this morning for the family $\lambda = (2, 2, 1^m)$, leaving the merged-arc (broken-junction) reach as the sole remaining gap. We give a sharper reformulation: *the lower bound is an operator (not trace) statement*. For every shape tested, $\min\{|S| : \text{tr}_{V^\lambda} \prod_k A_k(S) \neq 0\}$ equals $\min\{|S| : \prod_k A_k(S) \neq 0 \text{ as an operator on } V^\lambda\}$, and the trace order law reduces to the operator one. Reading the staircase product from right to left, every factor is a projector, so the running rank is monotone non-increasing; the operator vanishes iff this running rank reaches 0. Three new structural facts emerge: (i) the first far-junction crossing with both positions un-inserted always pinches the running rank to ≤ 1 (a clean operator lift of the pair-pinch character identity); (ii) every critical survivor at $\lambda = (2, 2, 1, 1, 1)$ becomes rank-1 at the first such crossing and *remains* rank-1 to the end; (iii) the within-arc reach lemma lifts verbatim to the operator level, giving a uniform proof of the lower bound $\text{ord} \geq \binom{m}{2}$ for *intact* subsets. The merged-arc obstruction narrows to a clean rank-1 vector-flow reach: the cyclic-trace factorisation is gone, but a different obstruction remains — the inert-top restriction breaks precisely when the super-arc generator sequence contains a generator adjacent to the surviving inert top. We delimit the gap precisely and indicate the route most consistent with the new framing.

1 Setup and the operator reformulation

We adopt the conventions of the two predecessor papers `2026-05-26-projector-grading-vanishing.tex` (hooks), `2026-05-27-order-law-offhook.tex`, and `2026-05-28-offhook-within-arc-reach.tex`.

The Hecke algebra $H_n(q)$ acts on the Hoefsmit seminormal module V^λ ; $P_q^{(i)} = (T_i + 1)/(q + 1)$ and $P_{-1}^{(i)} = (q - T_i)/(q + 1)$ are the complementary q - and (-1) -eigenprojectors of T_i on V^λ . Write $W_i := \text{im } P_q^{(i)}$ and $W_i^- := \text{im } P_{-1}^{(i)}$.

Via the Möbius substitution $s = \frac{q^2 - x}{q(x-1)}$,

$$\text{ord}_{x=q^2} Z_\lambda = \text{ord}_{s=0} P_\lambda(s), \quad P_\lambda(s) = \text{tr}_{V^\lambda} \prod_{k=0}^{N-1} (P_q^{(i_k)} + s P_{-1}^{(i_k)}) = \sum_S s^{|S|} c_S,$$

$N = \binom{n}{2}$, the product is over the staircase word $w = B_2 B_3 \cdots B_n$, $B_k = (k-1, k-2, \dots, 1)$, and

$$c_S = \text{tr}_{V^\lambda} M(S), \quad M(S) := \prod_{k=0}^{N-1} A_k(S), \quad A_k(S) = \begin{cases} P_{-1}^{(i_k)}, & k \in S, \\ P_q^{(i_k)}, & k \notin S. \end{cases}$$

$M(S)$ is an *operator* on V^λ ; c_S is its trace.

Theorem 1 (Operator reformulation). *Let*

$$g_\lambda^{\text{op}} := \min\{|S| : M(S) \neq 0 \text{ as operator on } V^\lambda\}, \quad g_\lambda^{\text{tr}} := \min\{|S| : c_S \neq 0\}.$$

Then $g_\lambda^{\text{op}} \leq g_\lambda^{\text{tr}} \leq \text{ord}_{s=0} P_\lambda$. For every shape verified ($\lambda \in \{(2, 1^m)_{m \leq 4}, (2, 2, 1^m)_{m \leq 3}, (2, 1, 1, 1)\}$ in this session), $g_\lambda^{\text{op}} = g_\lambda^{\text{tr}} = \text{ord}_{s=0} P_\lambda$, so the order law reduces to the operator min-support.

Proof. The first inequality is immediate: if $M(S) = 0$ then $\text{tr } M(S) = c_S = 0$, so operator vanishing on $\{|S| < g_\lambda^{\text{op}}\}$ forces $c_S = 0$ there. The second inequality is the standard $G_m = \sum_{|S|=m} c_S$: $c_S = 0$ for $|S| < m$ gives $G_m = 0$, so the polynomial order is at least g_λ^{tr} . Equality for the relevant shapes was verified directly: see `probe_op_universal.py`. $\square$

The point of Theorem ?? is that *the lower bound is purely an operator (rank) statement*: to show $\text{ord} \geq \binom{\tau+1}{2}$ it suffices to show $M(S) = 0$ for all S with $|S| < \binom{\tau+1}{2}$, with no traces or cyclic factorisation. Consequently *the intact/merged dichotomy that organised the trace approach dissolves at operator level* — $M(S)$ is one product, not a cyclic trace.

2 Monotone rank and the running-image flow

Lemma 2 (Monotone rank). *For any S and any $0 \leq i \leq j \leq N - 1$, $\text{rank}(\prod_{k=i}^j A_k(S))$ is non-increasing as i decreases (equivalently, reading the product right-to-left, the rank of the partial product is non-increasing as more factors are accumulated). In particular $M(S) \neq 0$ iff this running rank reaches the leftmost factor still positive.*

Proof. Every $A_k(S)$ is a projector. For any operator X and projector A , $\text{rank}(AX) \leq \min(\text{rank } A, \text{rank } X) \leq \text{rank } X$. So left-multiplying by A_k weakly decreases rank. $\square$

We henceforth read the staircase product right-to-left. The running image $I_k := \text{im}(\prod_{j \geq k} A_j(S))$ is a nested non-increasing sequence of subspaces of V^λ ; $M(S) = 0$ iff $I_0 = \{0\}$.

3 Mixed pinch dimensions

Lemma 3 (Mixed pair pinch). *Let $\lambda = (2, 2, 1^m)$ and let a, b be Hecke generator indices with $|a - b| \geq 2$. For each choice of signs $\varepsilon, \delta \in \{+, -\}$, the joint eigenspace $W_a^\varepsilon \cap W_b^\delta$ has dimension*

$$\dim(W_a^\varepsilon \cap W_b^\delta) = \frac{1}{4}(f^\lambda + \varepsilon \chi_{(2, 1^{n-2})}^\lambda + \delta \chi_{(2, 1^{n-2})}^\lambda + \varepsilon \delta \chi_{(2, 2, 1^{n-4})}^\lambda).$$

For all $(2, 2, 1^m)$ with $m \geq 1$ and far a, b , the four dimensions are $(1, 3, 3, 7)$ for signs $(++, +-, -+, --)$ in the case $n = 7$ ($\lambda = (2, 2, 1, 1, 1)$); the analogous fixed multiplet for general m arises from the same character formula.

Proof. T_a and T_b commute when $|a - b| \geq 2$, so V^λ decomposes as a direct sum of joint eigenspaces $W_a^\varepsilon \cap W_b^\delta$. At $q = 1$, $T_i \mapsto s_i$ and W_i^+, W_i^- are the ± 1 -eigenspaces of s_i . Joint eigenspace dimensions are deformation invariant, so we compute at $q = 1$. Each joint ± 1 -eigenspace of $\langle s_a \rangle \times \langle s_b \rangle \cong (\mathbb{Z}/2)^2$ has dimension $\frac{1}{4} \sum_{(\eta, \zeta) \in \{\pm 1\}^2} \eta^{(1-\varepsilon)/2} \zeta^{(1-\delta)/2} \chi^\lambda(s_a^{(1-\eta)/2} s_b^{(1-\zeta)/2})$ by character orthogonality. As a, b are non-adjacent, $s_a s_b$ has cycle type $(2, 2, 1^{n-4})$ and s_a, s_b each have cycle type $(2, 1^{n-2})$. The numerical values for $\lambda = (2, 2, 1^m)$ follow from Murnaghan–Nakayama; for $n = 7$ one obtains $(1, 3, 3, 7)$. $\square$

Remark 4. The same character calculation with three pairwise-non-adjacent generators $1, b, c$ gives $\dim(W_1 \cap W_b \cap W_c) = \frac{1}{8}(f + 3\chi_{(2)} + 3\chi_{(2,2)} + \chi_{(2,2,2)}) = 0$ for $\lambda = (2, 2, 1^m)$ ($m \geq 2$): the triple pinch identity of the morning paper.

4 The first–pinch collapse

The staircase word $w = B_2 B_3 \cdots B_n$ has off–band cyclic adjacencies (*far junctions*) only at positions $(p_k, p_k + 1)$ where $i_{p_k} = 1$ is the bottom of block B_k and $i_{p_k+1} = k$ is the top of B_{k+1} , for $k = 3, \dots, n - 1$. Within each block, consecutive generators differ by 1 (adjacent).

Theorem 5 (First–pinch operator collapse). *Let $S \subseteq [0, N - 1]$ and let p be the rightmost position of w such that $p \notin S$, $p + 1 \notin S$, and $(p, p + 1)$ is a far–junction adjacency ($|i_p - i_{p+1}| \geq 2$). Then after processing positions $p, p + 1, \dots, N - 1$ in the right–to–left rank flow of $M(S)$, the running rank is at most 1 and the running image is contained in $W_1 \cap W_{i_{p+1}}$ (of dimension 1 by Lemma ??).*

Proof. Let $p + 1 \leq k \leq N - 1$ be the positions strictly above p . Each $A_k(S)$ ($k > p + 1$) is a projector at a generator i_k adjacent to i_{p+1} (within the same block as i_{p+1} , since p is the rightmost far–junction position). All these factors commute with $P_q^{(i_{p+1})}$ when their generators are $\geq |i_{p+1} - i_k| \geq 2$, but actually we only need that the leftmost factor of block B_k for the block containing position $p + 1$ is i_{p+1} itself (the top generator). Specifically, $A_{p+1}(S)$ is a projector onto $W_{i_{p+1}}$ or $W_{i_{p+1}}^-$ depending on whether $p + 1 \in S$; by hypothesis $p + 1 \notin S$ so $A_{p+1}(S) = P_q^{(i_{p+1})}$ and $\text{im}(\prod_{j \geq p+1} A_j) \subseteq W_{i_{p+1}}$.

Next apply $A_p(S) = P_q^{(1)}$ (since $p \notin S$). The composition $P_q^{(1)} P_q^{(i_{p+1})}$ is, by Lemma ?? with $a = 1, b = i_{p+1}, \varepsilon = \delta = +$, a projector onto $W_1 \cap W_{i_{p+1}}$, which has dimension 1. Hence $\text{rank}(P_q^{(1)} P_q^{(i_{p+1})}) = 1$. Since the running image after position $p + 1$ is contained in $W_{i_{p+1}}$, applying $P_q^{(1)}$ now maps it into $W_1 \cap W_{i_{p+1}}$, and the resulting rank is at most 1. $\square$

Corollary 6 (Rank–1 regime for survivors at $\lambda = (2, 2, 1, 1, 1)$). *Every S with $M(S) \neq 0$ at $\lambda = (2, 2, 1, 1, 1)$ has a first far–junction position $p \notin S, p + 1 \notin S$ in the right–to–left flow: the running rank is 1 from position p onward all the way to position 0.*

Proof. There are four far junctions for $n = 7$, at $(2, 3), (5, 6), (9, 10), (14, 15)$. Any S with $|S| \leq 3$ contains at most 3 of the 8 far–junction positions $\{2, 3, 5, 6, 9, 10, 14, 15\}$; at most three junctions can be broken; so at least one is intact and one is the rightmost intact junction p . Applying Theorem ??, the running rank is ≤ 1 from p leftward; since $M(S) \neq 0$ requires running rank ≥ 1 everywhere left of p , it is exactly 1 throughout. This was verified explicitly for all 14 critical survivors of $(2, 2, 1, 1, 1)$ (script `probe_gates.py`). $\square$

5 Operator lift of the within–arc reach

The morning paper proved the within–arc reach lemma for $\lambda = (2, 2, 1^m)$ at *trace* level: in the cyclic trace factorisation, the arc–scalar at a zero arc with parameter a vanishes unless the arc subset contains $\{3, \dots, a\}$. The proof used (i) restriction to the inert top W_{a+2} , restoring rank–one projectors on a subspace, and (ii) the triple pinch sign–kill on the left covector.

Theorem 7 (Operator within–arc reach). *Let S be a subset with all far–junction positions $\notin S$ (intact case). For each zero arc with parameter $a \geq 3$ (between the far junctions $(1, a - 1)$ and $(1, a + 1)$), the running rank–1 vector flow through the arc generators $T_2, \dots, T_a$ vanishes unless the arc positions corresponding to generators $\{3, \dots, a\}$ are all in S . In particular, the within–arc cost is exactly $a - 2$, uniquely attained at the descending–run prefix.*

Proof. By Theorem ?? (applied iteratively as each intact far junction is crossed right–to–left), after the first crossing the running rank is 1 and the running image lives in $W_1 \cap W_{i_{p+1}}$. As the flow continues left through the arc, within each block B_k the inert top T_{k+1} (next block’s top,

processed immediately before in R-to-L order) commutes with all generators $T_2, \dots, T_{k-1}$ inside B_k . Restricting the flow to $W = W_{k+1}$ (which is preserved by every arc generator), each $P_q^{(j)}|_W$ ($2 \leq j \leq k-1$) is rank one with image $W_j \cap W_{k+1}$ (dim 1 by pair pinch).

The operator version of the May 28 “walk-expansion” argument now applies verbatim: the running rank-1 vector $v_{\text{in}} \in W_1 \cap W_{k+1}$ is carried through the rank-one operators A_j on W , and the resulting vector at the next crossing (to which $P_q^{(1)}$ will be applied) is $y_{\text{eff}}^T \cdot (\prod_{j \leq a} A_j) \cdot p_1$ *exactly* as in the trace-level scalar (Eq. (??) of the morning paper). The triple-pinch sign-kill gives $y_{\text{eff}}^T p_j = 0$ for $j \geq 3$, forcing the walk to remain on vertices $\{1, 2\}$; this is possible only if $\{3, \dots, a\} \subseteq S$, with unique minimiser the descending prefix. The trace-level argument is the scalar shadow of an operator-level vector flow on W . $\square$

Corollary 8 (Intact-junction lower bound, operator version). *For $\lambda = (2, 2, 1^m)$ and any S with no far-junction position in S , if $|S| < \binom{m}{2}$ then $M(S) = 0$.*

Proof. $|S| = \sum_a |S \cap (\text{zero arc with parameter } a)|$. By Theorem ??, $M(S) \neq 0$ requires each zero arc to receive at least $a - 2$ insertions; summing over $a = 3, \dots, n - 3$ gives $|S| \geq \binom{m}{2}$. $\square$

6 The merged-junction gap, sharpened

When S places a P_{-1} at a far-junction position, two adjacent arcs merge into a super-arc. At trace level this destroys the cyclic factorisation. At operator level Theorem ?? still gives a rank-1 regime after the first *intact* crossing, but Theorem ??’s inert-top restriction now fails: the merged super-arc may contain a generator adjacent to the surviving inert top.

Remark 9 (The obstruction, precisely). For $\lambda = (2, 2, 1, 1, 1)$, breaking the shared junction $(1, 5)$ at position 9 or 10 merges the zero arcs $a = 3$ (in B_5) and $a = 4$ (in B_6) into a super-arc with generator sequence $(3, 2, 1, 5, 4, 3, 2)$. The surviving inert top is T_6 (from the still-intact $(1, 6)$ junction); but the super-arc contains T_5 , adjacent to T_6 , so W_6 is *not* invariant under $P_q^{(5)}$. The rank-1 vector flow leaves the joint eigenspace exactly at this generator. Restricting to $W_5 \cap W_6 = \{0\}$ collapses spuriously (refuted this wake; see `glued-restriction.py`); restricting to W_1 also fails as a chip-firing host (only some generators compress to rank 1; see `W1-merged-chipfiring.py`).

Remark 10 (Empirical refutation of a candidate “empty ++-gap” criterion). A natural conjecture — if two far-junction crossings are both intact (++) and no insertion sits strictly between their gen-1 positions in the R-to-L order, then $M(S) = 0$ — is refuted: 8 of the 14 critical survivors at $(2, 2, 1, 1, 1)$ have such an empty ++-gap (verified, `probe_crux.py`). The reason is precisely that of Remark ??: between two “intact” crossings the carried rank-1 vector need not remain in the joint q -eigenspace of two far generators, so the sign-kill that would force it into W_1^- at the next intact crossing fails.

Remark 11 (The operator merged-reach lemma, conjectural). For a super-arc covering zero arcs with parameters $a_1 < \dots < a_p$, the running rank-1 vector flow has reach $\sum_i (a_i - 2)$: the super-arc requires at least $\sum_i (a_i - 2)$ insertions inside it (counting broken far-junction positions). At operator level this is purely a vector flow problem, with no cyclic-trace dependency. The candidate proof route that survives both this wake’s two refutations is a *sequential* use of inert tops — one per merged block — with a careful tracking of how the running vector transfers across each broken junction. See `merged-junction-mechanism.md`.

7 What is proved, what remains

Proved (this session).

- Operator reformulation (Theorem ??): the order-law lower bound reduces to the operator min-support g_λ^{op} .
- Monotone rank flow (Lemma ??): a clean structural framework for the lower bound that is uniform in S (no intact/merged dichotomy at this level).
- Mixed pinch dimensions (Lemma ??): explicit signed joint-eigenspace dimensions via characters, generalising the pair pinch.
- First-pinch collapse (Theorem ??): the first intact far-junction crossing pinches the running rank to ≤ 1 unconditionally.
- Operator within-arc reach (Theorem ??): the morning's arc reach lifts to a rank-1 vector flow, giving the lower bound $\text{ord} \geq \binom{m}{2}$ for far-junction-intact subsets (Corollary ??).

Remaining (the gap). The merged-arc operator reach (Remarks ?? and following): a rank-1 vector flow through a super-arc that contains a generator adjacent to the surviving inert top. The operator framing removes the cyclic-trace obstruction (one gain over the trace-level formulation) but the inert-top non-invariance is genuine and is the precise structural difficulty. The next attempt should construct a *sequential* inert-top restriction (one per merged block, with a tracked transfer across the broken junction) rather than a single global restriction.

For $\lambda = (2, 2, 1, 1, 1)$. The order law $\text{ord} = 3$ is computationally unconditional (all $\binom{21}{\leq 2} = 232$ subsets checked: $M(S) = 0$ at operator level; the 14 critical survivors at $|S| = 3$ all give $c_S = q^8/(q+1)^{16} > 0$, so $\text{ord} = 3$ exactly). The operator framework reduces the structural lower bound to the merged-arc reach; for intact subsets the lower bound is now also unconditional via Corollary ??.

8 Verification

All claims tested in `sympy` at $q = 5$ and $q = 7/3$ on $\lambda \in \{(2, 1^m)_{m \leq 4}, (2, 2, 1^m)_{m \leq 3}, (2, 1, 1, 1)\}$. Scripts in `~/projects/scratch/2026-05-23-omega-monodromy-verify/` and `/tmp/`:

- `probe_op_universal.py`: $g_\lambda^{\text{op}} = g_\lambda^{\text{tr}}$ on all tested shapes (Theorem ??).
- `probe_operator_vanish.py`: $M(S) \equiv 0$ as operator for all $|S| \leq 2$ at $\lambda = (2, 2, 1, 1, 1)$.
- `probe_running_rank.py`, `probe_rank_ladder.py`: rank ladder $\text{rank}(\Omega) = m + 1 \rightarrow 1 \rightarrow 0$ at $\lambda = (2, 2, 1^m)$ for $m = 2, 3, 4$; universal rank-1 collapse at the first intact crossing for all 14 $(2, 2, 1, 1, 1)$ survivors (Corollary ??).
- `probe_transfer.py`: mixed pair pinch dimensions $(1, 3, 3, 7)$ for $\lambda = (2, 2, 1, 1, 1)$ uniform over far pairs (Lemma ??); triple pinch vanishing.
- `probe_crux.py`: refutation of the “empty ++-gap” criterion (8/14 survivors with empty gap).
- `probe_windows.py`: the contiguous-window count is too weak (can force ≥ 2 at most, not ≥ 3) — confirming that the graded reach is genuinely required.